Data, models and evaluation (`ml/`) — consolidated report

Generated 2026-10-19. Everything in this report was built in this project. Every number was written by a script that ran and was read back out of the file it produced — none is estimated.

Full per-model detail: `docs/dataset_and_eda.md` · `docs/model_card.md` · `docs/model_card_movement.md` · `docs/future_risk_prediction.md` · `docs/test_report.md` **Plain English:** `docs/data_ml_explained_simply.md`

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1. What was built

Component	Trained?	Method	Artifact	Runs where
Trajectory anomaly detector	Unsupervised	`IsolationForest`, 200 trees, 14 features	`ml/models/isolation_forest.joblib` (1,753.5 KB)	`ml/` analysis
Activity recognition	Yes	`RandomForest`, 32 trees, depth 8, 24 features	`app/src/main/assets/har_activity_model.json` (268,734 B)	**On the phone**
Impact (fall-signature) detector	Yes	`RandomForest`, 40 trees, 24 features	`app/src/main/assets/har_fall_model.json` (59,679 B)	**On the phone**
Future-risk predictor	Yes	`RandomForest`, 120 trees, depth 11, 22 features + Platt calibration	`app/src/main/assets/future_risk_model.json` (1,974,297 B), version `future-risk-unimib-v1`	**On the phone**
Transport-mode classifier	Yes	`RandomForest`, 300 trees, 9 features	`ml/outputs/transport/`	`ml/` analysis (not shipped)
Geofence	No — configured rule	Polygon test + event log	`app/` + `backend/`	Phone + server
Approach-risk score	No — heuristic	Published weighted checklist	`app/`	Phone

Three models are exported to JSON and walked by hand-written Kotlin **on the device**, with no TensorFlow, TFLite or PyTorch, and no network call at inference time. The two HAR models together are under 330 KB, which is why a hand-written walker was chosen over a runtime.

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2. Datasets used

2.1 Inventory

Dataset	What it actually is	Scale (measured)	Licence / terms	Used for
GeoLife GPS Trajectories 1.3	GPS traces from volunteers, 2000–2012, mostly Beijing	Archive 313,164,406 B; **182 users; 18,670 `.plt` files; 24,876,978 data rows; 0 malformed**	**No OSI licence bundled** — citation-only, research use	EDA, anomaly detection, transport model, zone analysis
UCI HAR	30 volunteers, scripted activities, waist-mounted phone	**7,352 train / 2,947 test windows**, 561 features, 6 activities, 9 channels @ 50 Hz	Not documented in this repository — see the UCI dataset page	EDA, activity recognition, impact detector
UniMiB SHAR	30 volunteers performing 9 daily activities and **8 real fall types** in a lab	Archive 266,206,080 B, SHA-256 `365a3967…67160`; 1,980 trials read, **1,979 used** @ 50 Hz	Publicly distributed for academic research; **no explicit OSI licence**	The future-risk model
OpenStreetMap (Overpass API)	Live public map database, bounded extract	**15,980 elements** in 1.5 km × 1.5 km of Tiananmen, Beijing	© OpenStreetMap contributors, **ODbL**	Map context EDA only — **no model input**
Open-Meteo weather archive	Historical weather for two sites	**574 days**, 2011-01-01…2012-07-27	Free/open archive	Context EDA only — **no model input**
Synthetic fixture	Project-generated stand-in	tiny	project-internal	Plumbing test when the real archive is absent

2.2 The facts that shape everything downstream

- **GeoLife contains no danger, injury or incident labels.** Its only labels are transportation modes (69 users, 14,718 segments). A supervised risk model is therefore impossible, and the trajectory model is **unsupervised** — its output is a movement *anomaly score*, never a probability of danger.
- **UCI HAR contains no falls at all** — no fall class, no impact annotation. The impact detector's positive class was therefore **manufactured** by injecting a fall-like transient into real held-out recordings. Every metric for that model measures detection of the **injected signature**.
- **UniMiB SHAR is the only dataset here with a real recorded outcome** (a real fall). That is why the future-risk model is the only one that predicts a *future event* rather than describing the present.
- **The weather provider actually used is Open-Meteo, not OpenWeatherMap.** The brief named OpenWeatherMap, which needs a key we did not have; the fallback is stated on every artefact.
- **Two corrections are recorded rather than hidden:** 698,900 duplicate `(trajectory, timestamp)` rows were removed from GeoLife, and **one corrupt UniMiB recording** (subject 10, `Jumping`, trial 1 — 9,369 non-finite values of 9,624) was **dropped and counted**.

2.3 Dataset → model mapping

Dataset	Anomaly model	Activity model	Impact detector	Future-risk	Transport	EDA	Map
GeoLife 1.3	✓ **train + test**	–	–	–	✓ **labels**	✓	✓
UCI HAR	–	✓ **train + test**	✓ **train + test**	–	–	✓	–
UniMiB SHAR	–	–	–	✓ **train + test**	–	✓	–
OpenStreetMap	–	–	–	–	–	✓	✓
Open-Meteo	–	–	–	–	–	✓	–

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3. EDA performed

3.1 GeoLife — the twelve analyses

Point-level figures use a fixed-seed sample of **1,429,740 points / 1,200 trajectories**; length, duration and per-user figures use all 18,670 trajectories. Outputs: `ml/outputs/eda/` (12 figures, 11 tables, `findings.md`).

#	Analysis	Measured result
1	Shape / dtypes	1,429,740 × 15 (point sample); 18,670 × 22 (trajectory summary)
2	Missing / duplicates	698,900 duplicate rows, 1 invalid coordinate, 0 unparsable, 0 malformed; `dt_s` missing = 1,200 (structural)
3	Spatial extent	lat 1.27681…51.10710, lon −179.69134…179.99330; density overwhelmingly Beijing
4	Length / duration	points median **504**, mean 1,295, max 92,645; duration median **2,528 s**, max 348,431 s; total distance **1,021,454 km**
5	Sampling interval	median `dt` **2.00 s**, mean 4.74, p95 5.00, max 3,595; **96.3 %** in the 1–5 s band
6	Speed	4,525 of 1,423,352 steps > 60 m/s **excluded**; speed mean **6.55 m/s**, median 3.23, p95 20.72
7	Dwell / stops	**557** stops (speed < 0.5 m/s for ≥ 120 s); median 174 s, max 9,945 s
8	Time of day	busiest **09:00** (108,135 fixes); quietest **21:00** (11,095); **no timezone stored**
9	Spatial plot	60 sample trajectories
10	Outliers	`alt_change` 47.55 % flagged, `straightness_w5` 14.86 %, `accel_mps2` 13.45 %, `turn_deg` 12.26 %, `altitude` 11.21 %
11	Correlation	strongest pair `speed_mps ~ speed_mean_w5`, **r = 0.979** (same motion, different window)
12	Class balance	14,718 transport labels across 69 users: walk 6,460 · bus 2,853 · bike 2,089 · taxi 1,179 · car 993 · subway 813 · train 299 · airplane 17 · boat 7 · run 6 · motorcycle 2. **No danger labels exist.**

Critical data quirk found: `.plt` file names **collide across users** — 847 ids are shared. Every grouping therefore uses `traj_uid` (user id + file stem); grouping on the bare file name would silently merge different people's trajectories, a leakage bug.

3.2 UCI HAR

- **Shape:** 561 features per window; 7,352 train / 2,947 test windows; 6 activities; raw window 128 samples @ 50 Hz (2.56 s), 9 inertial channels.
- **Split:** 21 train / 9 test subjects, **overlap 0** — genuinely subject-disjoint.
- **Class balance:** near-uniform (train 986–1,407; test 420–537 per activity). Accuracy is therefore a legitimate headline; no resampling or class weighting was needed.

- **Gravity sanity check:** mean total-acceleration magnitude **1.0323 g** (per-activity 1.0075–1.0516 g) — a real measurement centring on gravity. This is what justifies converting live m/s² samples to g.
- **Signal bands:** the three walking variants overlap heavily; sitting/standing/lying separate cleanly. This predicted the eventual confusion pattern exactly.
- **Feature redundancy:** the 561 features are heavily correlated; accuracy flattens past ~50 features.

3.3 UniMiB SHAR — the window and label construction

Quantity	Measured
Trials in / kept	1,980 / **1,979** (1 corrupt recording dropped)
Fall trials used	1,440 (6 excluded — never reached the 2.0 g impact threshold)
ADL trials	539
Windows cut	**51,128** (50-sample window, 10-sample stride)
Positives	**6,458** → positive rate **12.63 %**
Impact peak range	**2.0299 g … 3.4562 g**
Lead-time histogram (windows per lead)	50: 923 · 40: 1,268 · 30: 1,412 · 20: 1,425 · 10: 1,430

Why the horizon is one second — measured, not chosen: fall trials run 151–446 samples (median 174 = 3.5 s) with the impact near index ~120, leaving ≥ 95 samples of pre-impact signal. A 50-sample window ending at least one sample before impact fits in **every** included trial, so 1.0 s is the **longest fully supported** horizon.

Class imbalance is real and severe (12.63 % positive). Hence `class_weight="balanced_subsample"`, and precision / recall / PR-AUC reported alongside accuracy everywhere — a "always no" model already scores 87.1 % accuracy, and the PR-AUC baseline (0.128565) *is* the positive rate.

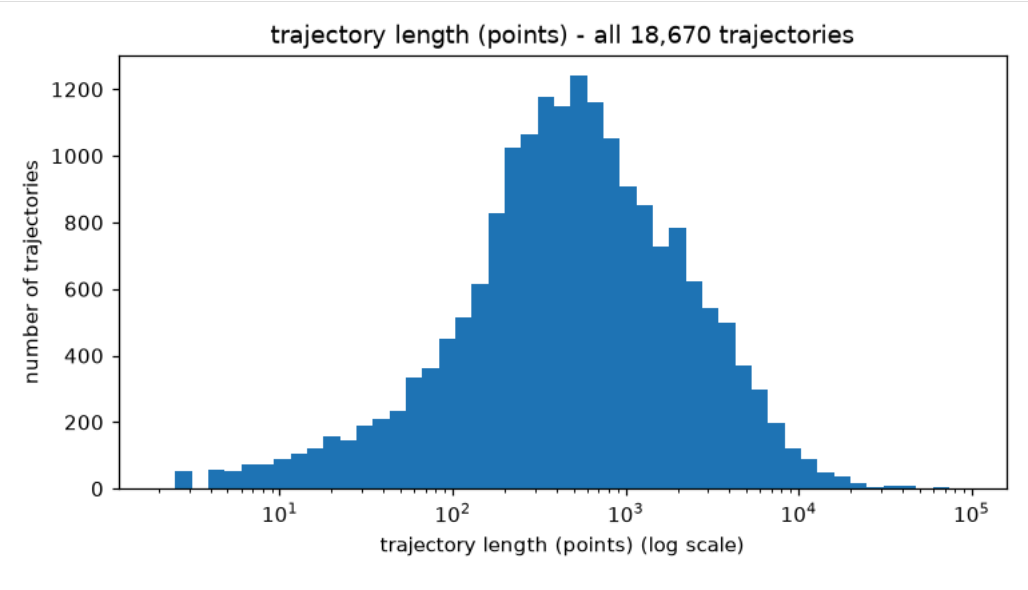
3.4 Map, weather and historical trends

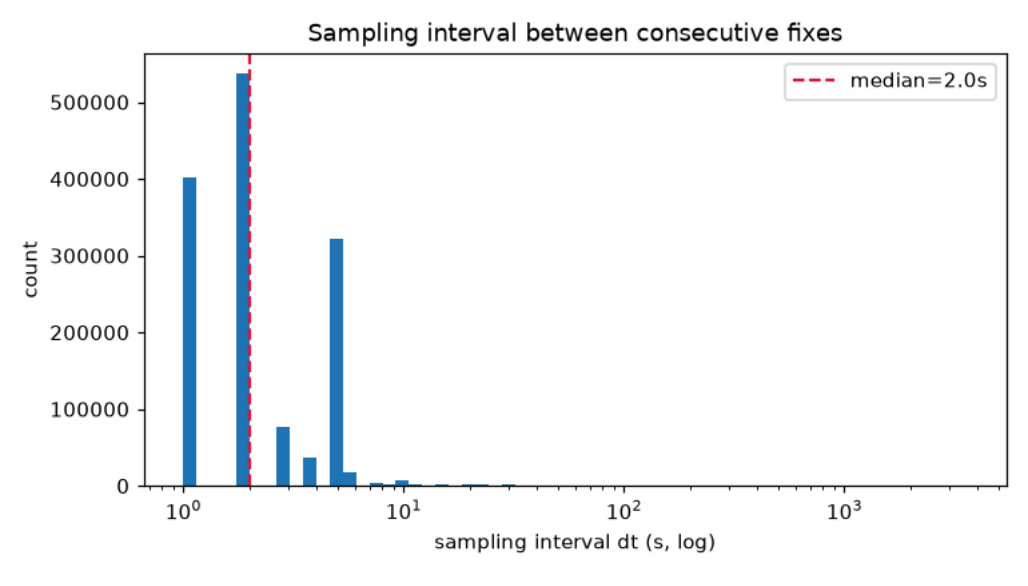
- **OpenStreetMap** (`ml/outputs/osm/`): 12,783 nodes, 3,197 ways, 0 relations = **15,980 elements**, **5,681.4 nodes/km²**. Top tags: `building` 1,205 · `highway` 1,075 · `height` 800. Highway classes: footway 504, pedestrian 114, steps 104 (17 distinct). Amenity classes: restaurant 36, toilets 22, café 12 (22 distinct). Fetched 2026-10-17T13:39:46Z, cached, and reproducible from the recorded query.
- **Weather** (`ml/outputs/weather/`): 574 days, two sites. `geolife_centroid` (39.9770 N, 116.3501 E, the **component-wise median** of the cleaned sample) → mean 12.17 °C, precip 806.4 mm, 175 harsh days (30.5 %). `huangshan_trek` (30.134 N, 118.161 E, an analyst-chosen trekking point) → mean 7.07 °C, precip 2,676.3 mm, 218 harsh days (38.0 %). Harsh-day rule: precip ≥ 10 mm **or** min T ≤ −5 °C **or** max T ≥ 35 °C **or** mean wind ≥ 30 km/h.
- **Historical / zone trends** (`ml/outputs/history/`): 18,670 trajectories / 182 users spanning 2000-01-01…2012-07-27. Trajectories per user median 28, max 2,153. Busiest hour 09:00, busiest weekday Saturday, peak month 2008-11. **557 stops, revisit rate 0.487**, most-revisited single place 19 stops. A zone derived from data density (densest 0.02° cell 39.9900 N/116.3300 E, expanded ±0.03° → S 39.95, W 116.29, N 40.03, E 116.37) is crossed by **431** of the 785 intersecting trajectories: **430 entries / 426 exits over 58 months**.

3.5 What the EDA changed

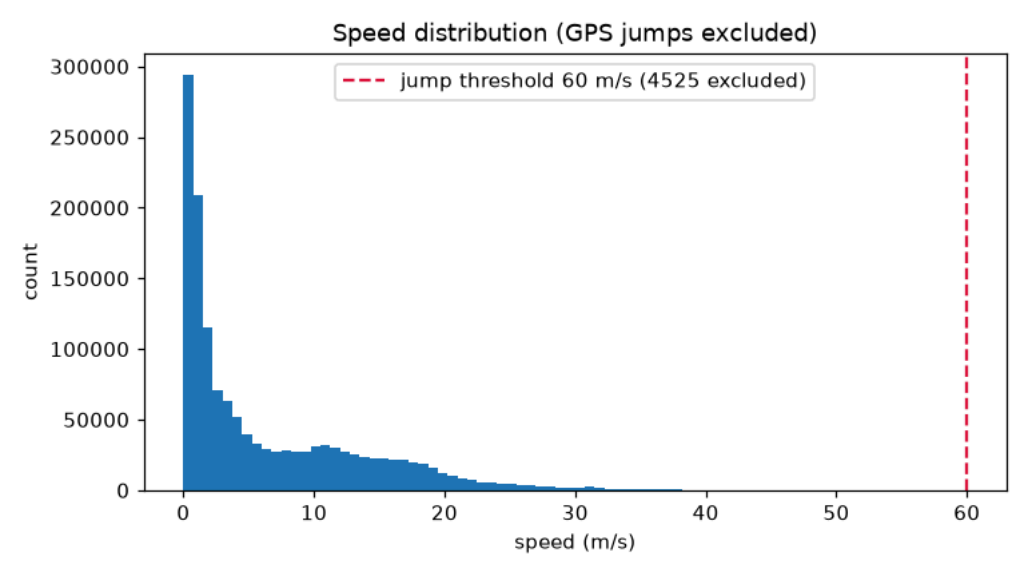
Finding	Consequent change
`.plt` names collide across users (847)	composite `traj_uid` grouping — prevents silent cross-user leakage
Irregular, gap-ridden sampling (1–3,595 s)	120 s segment split; `log_dt_s` as a feature; no resampling
GPS jumps (4,525 speeds > 60 m/s)	jumps kept as a data property; the evaluation injects teleports of the same family
`altitude` unreliable (47.55 % IQR-flagged)	documented as unreliable; no conclusion depends on it
Structural missingness (1,200 `dt_s`)	first point of each segment dropped, never imputed
HAR classes near-balanced	accuracy used as the headline; no resampling
HAR walking variants overlap	the app groups the three as "ambulating" and never acts on the distinction
561 HAR features redundant	24-feature on-device subset, with its measured 0.0936 accuracy cost reported
UniMiB 12.63 % positive rate	balanced class weights; precision/recall/PR-AUC reported everywhere
UniMiB impact always ≥ 2.0 g	2.0 g reused as the shared "hard landing" threshold in code and label rule
GeoLife has no danger labels	the trajectory model is unsupervised and never called a probability

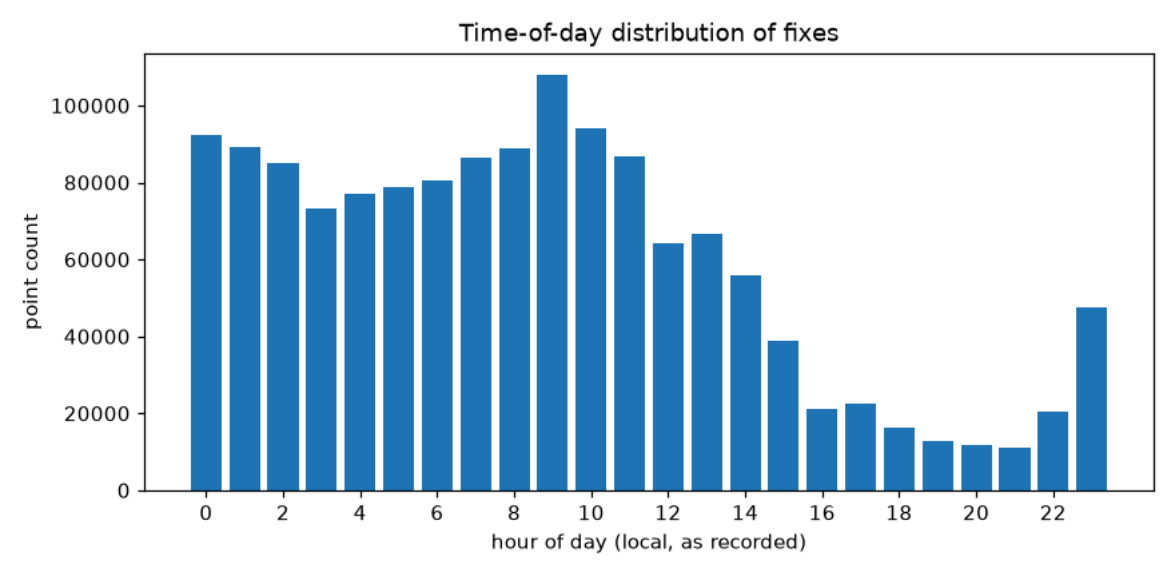
3.6 Figures from the EDA



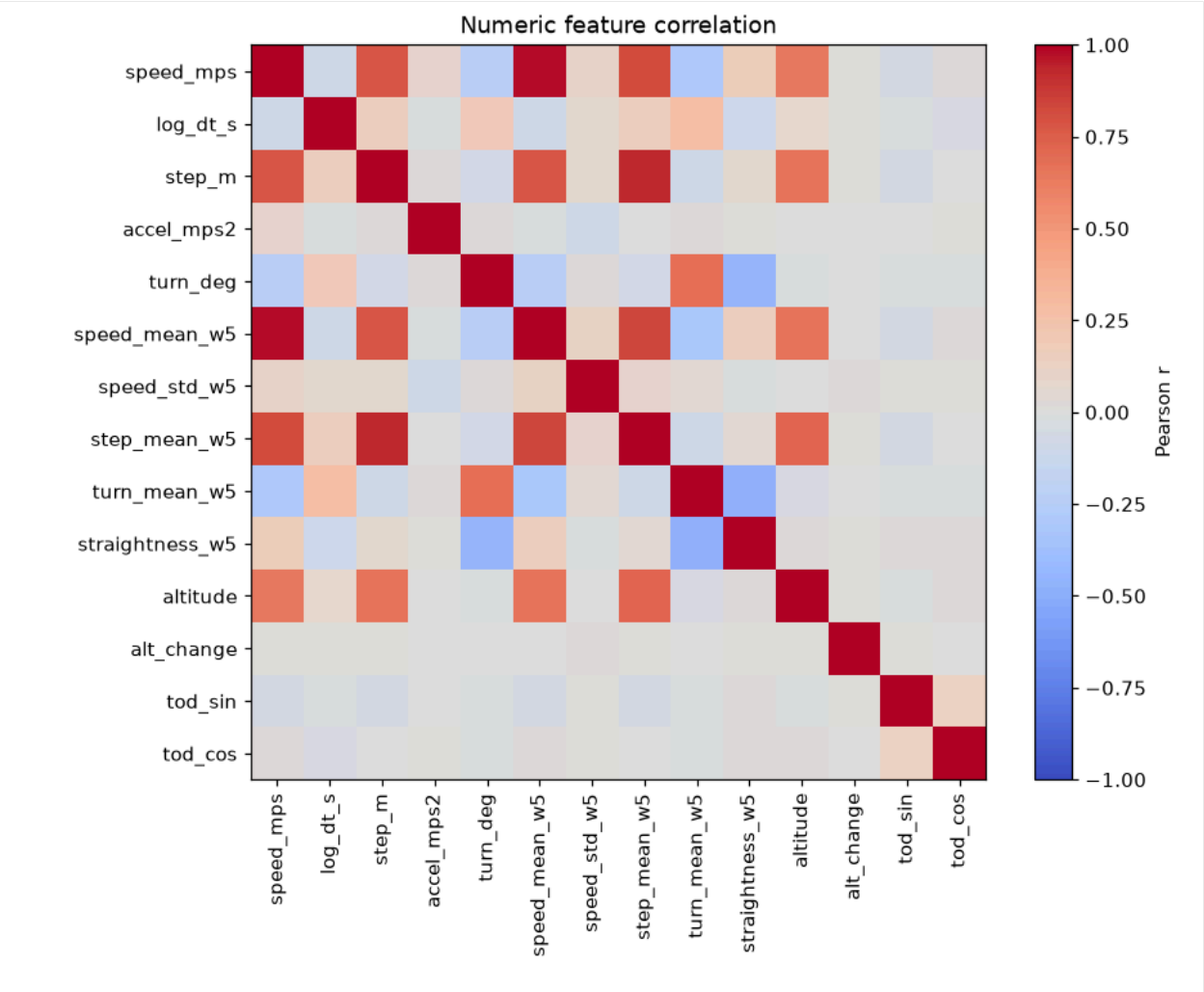


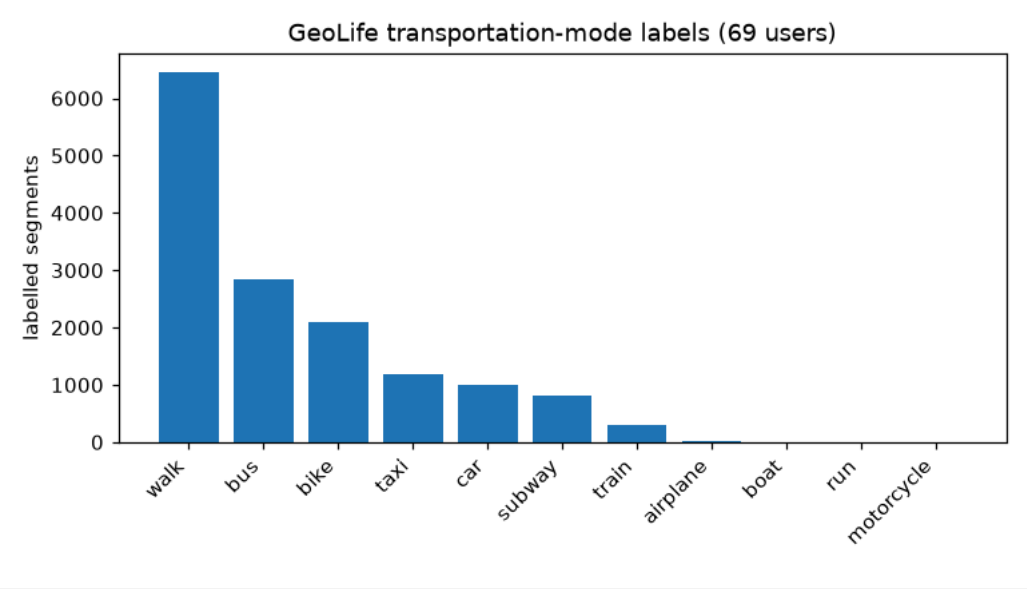
Length (median 504 points, heavy right tail) and sampling interval (96.3 % of gaps in 1–5 s). Both drove the choice of windowed features and a gap-based segment split rather than fixed-rate resampling.



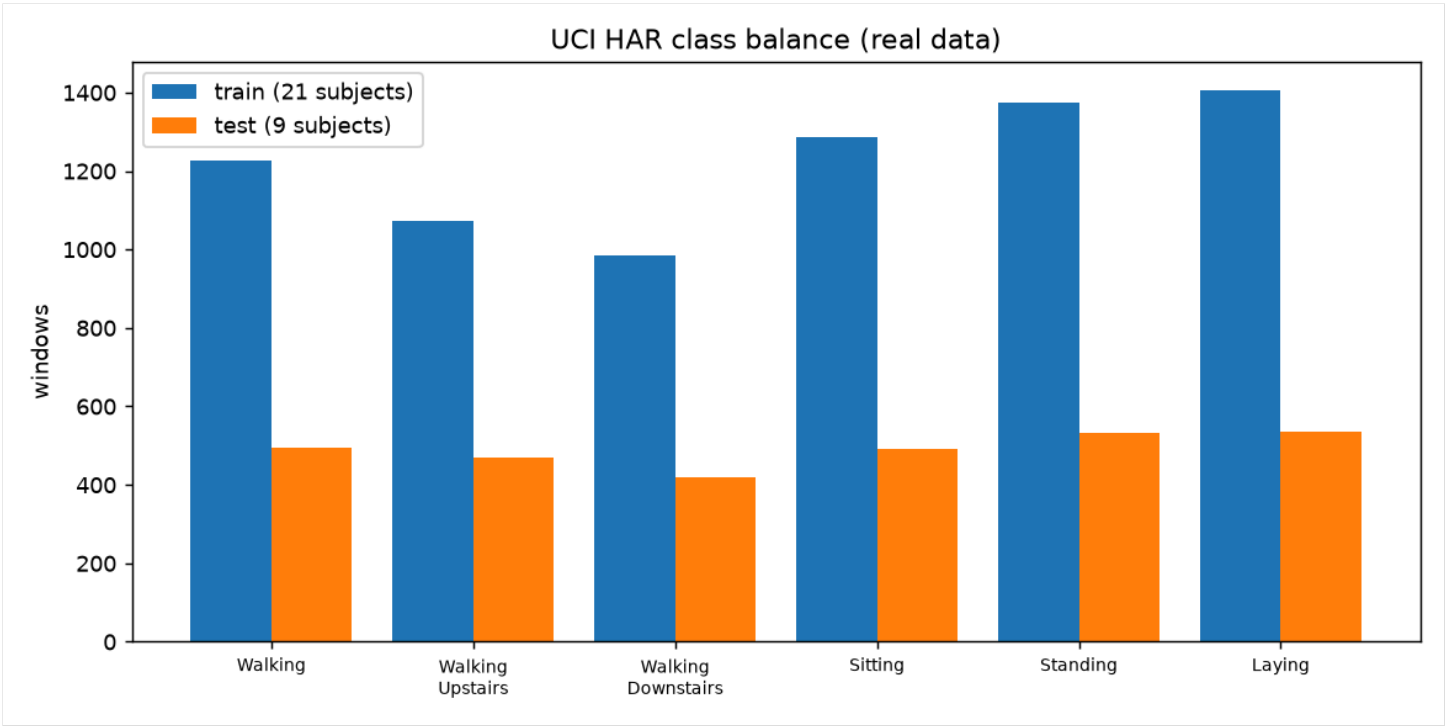


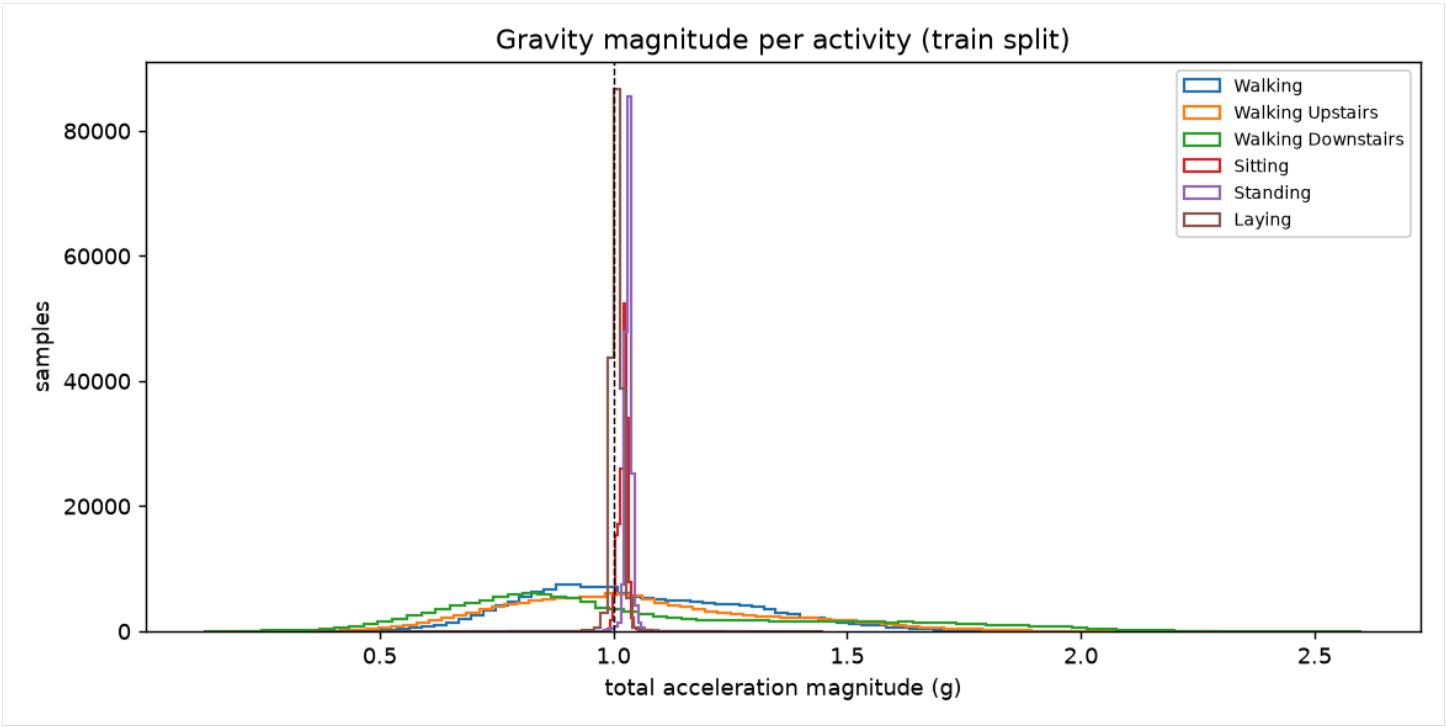
Speed after removing 4,525 impossible jumps (mean 6.55, median 3.23 m/s) — a plausible human range. The daily rhythm peaks at 09:00, which is why `tod_sin` / `tod_cos` *are model features.*



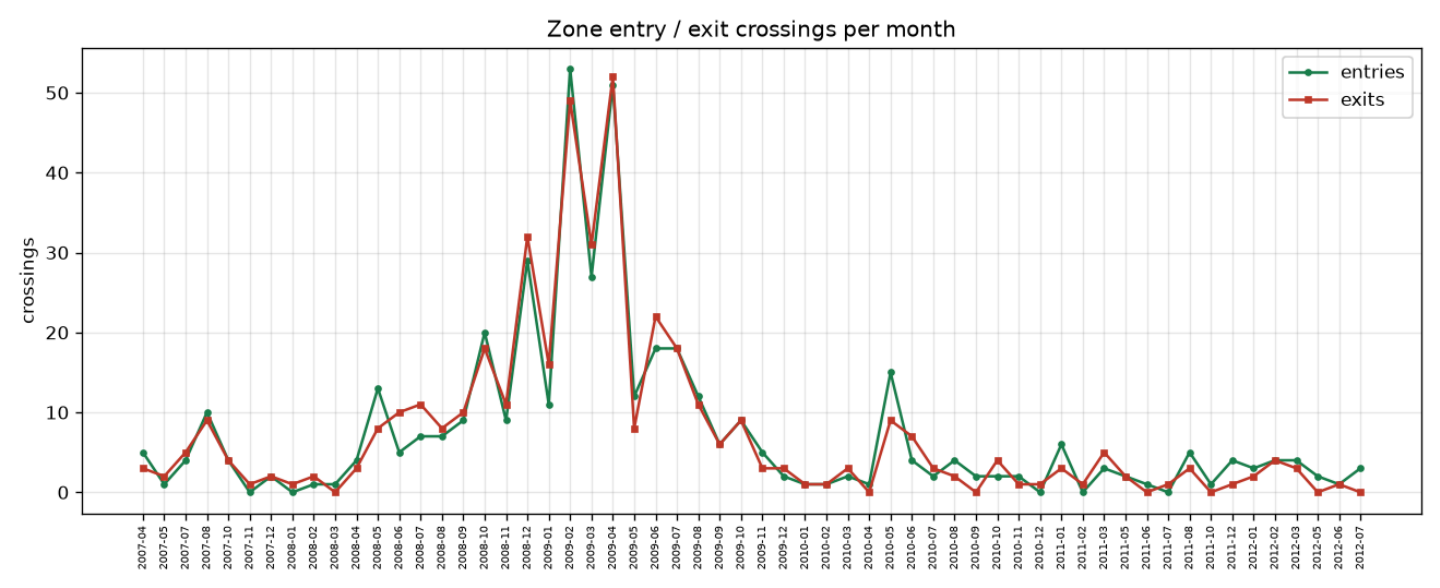


The strongest correlation is `speed_mps ~ speed_mean_w5` at r = 0.979 (harmless for an unsupervised scorer, documented rather than hidden). Transport labels are severely imbalanced — walk 6,460 down to motorcycle 2 — which is why the transport model uses balanced class weights.





UCI HAR's near-uniform classes justify accuracy as a headline. The gravity magnitude centres on 1.0323 g — a real accelerometer reading gravity, which is what justifies the m/s² → g conversion.



430 entries / 426 exits over 58 months in a data-derived Beijing zone: real evidence that boundary crossings happen at a countable rate.

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4. The models and what they measured

4.1 Trajectory anomaly detector (GeoLife, unsupervised)

`IsolationForest`, 200 trees, contamination 0.02, seed 42, over 14 standardised movement features. Score = `-score_samples(scaled_features)`; **higher = more anomalous, not a probability**. Trained on 1,060,493 points (900 trajectories), evaluated on 369,247 points (300 trajectories), split by `traj_uid` with **0 overlap**.

Accuracy, F1 and a confusion matrix do not exist for this model, and can't: GeoLife has no anomaly labels, so there is nothing to score against. Any such figure would be invented.

So it was validated by **injection on real data**: GPS teleport bursts (0.02°, run length 3) planted into real held-out trajectories.

Quantity	Value
Points marked affected	17,462
ROC-AUC	**0.8922**
Average precision (PR-AUC)	**0.6763**
Precision@k / Recall@k (k = 17,462)	**0.6165** / **0.6165**
Label-shuffle control ROC-AUC	**0.5007** (chance — the metric is meaningful)

Thresholds are score quantiles, not tuned: 0.98 → **0.5299** → **2.00 %** flagged (0.90 → 0.4588 → 10 %; 0.95 → 0.4853 → 5 %; 0.99 → 0.5810 → 1 %; 0.995 → 0.6374 → 0.5 %). Score distribution: train mean 0.4141 / std 0.0479, test mean 0.4080 / std 0.0437 — train and test match, so the model is not memorising the training split.

4.2 Activity recognition (UCI HAR)

Official subject-disjoint split: 7,352 train / 2,947 test windows, 21 / 9 subjects, overlap 0.

Model	Features	Trees	Accuracy	Macro-F1
Reference (full detail)	561	300	0.9287	0.9271
On-device (shipped)	**24**	**32**	**0.8351**	**0.8333**
Single decision tree (baseline)	24	1	0.7967	0.7943

Per-class, shipped 24-feature model:

Activity	Precision	Recall	F1	Support
WALKING	0.7363	0.8669	0.7963	496
WALKING_UPSTAIRS	0.7380	0.7176	0.7277	471
WALKING_DOWNSTAIRS	0.9507	0.7810	0.8575	420
SITTING	0.7996	0.8045	0.8020	491
STANDING	0.8185	0.8139	0.8162	532
LAYING	1.0000	1.0000	1.0000	537
Macro average	**0.8405**	**0.8307**	**0.8333**	**2,947**

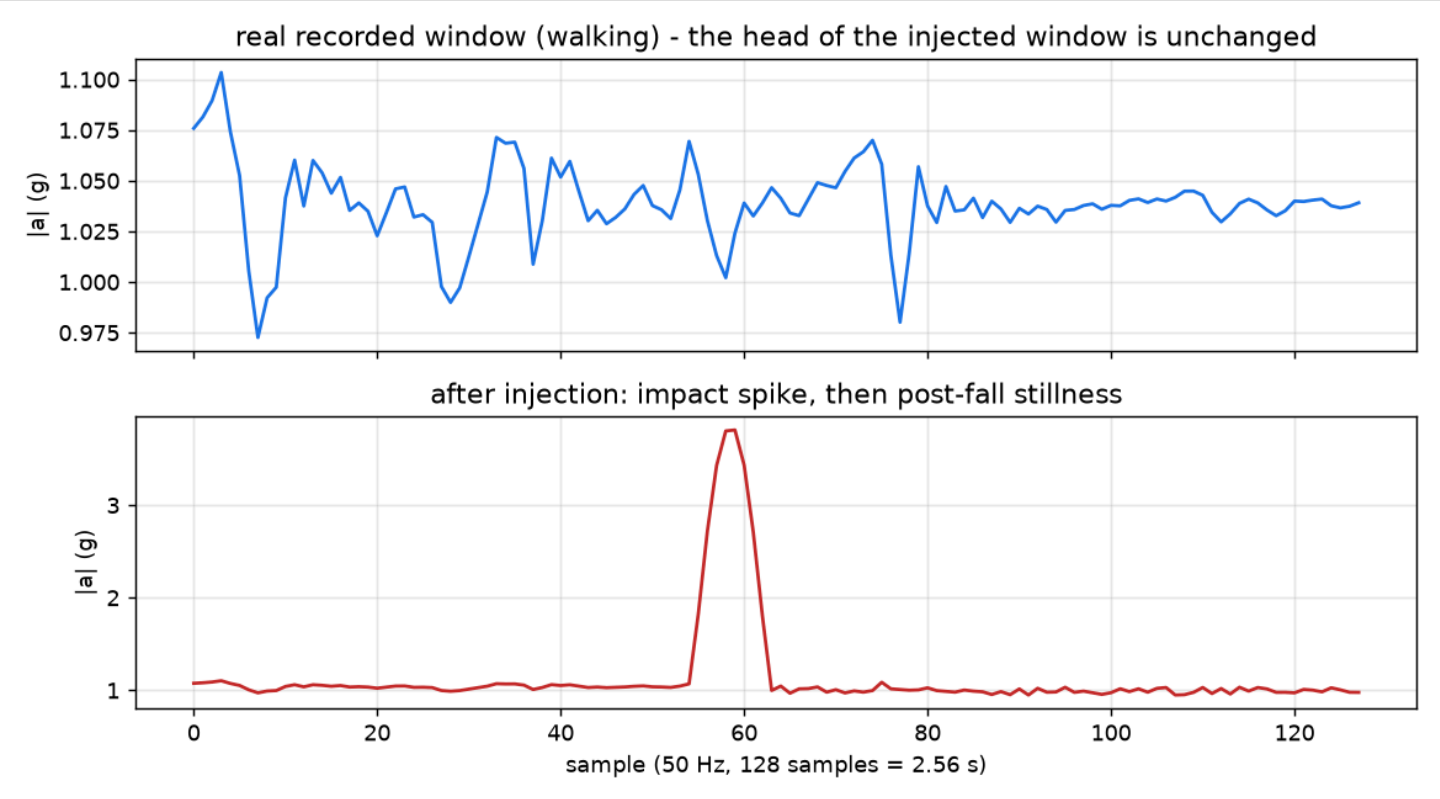
The cost of shipping the small model is **0.0936 accuracy**; the benefit over a single tree is **0.0384**. Both are reported because either alone misleads. The failures are concentrated in the three walking variants — which the app groups as one "ambulating" state and never acts on separately.

4.3 Impact (fall-signature) detector

Metric	Value
Test windows	1,848 (924 positive / 924 negative), subject-disjoint
ROC-AUC	**0.9999**
Average precision (PR-AUC)	**0.9999**
F1 / precision / recall @ 0.5	**0.9962** / 0.9957 / 0.9968
Shuffled-label control ROC-AUC	**0.5198**

Confusion matrix — derived from the saved precision/recall and n = 924 per class: **TP 921, FP 4, FN 3, TN 920** (accuracy 0.9962, specificity 0.9957).

How to read this. The shuffled-label control collapses to chance, so the near-perfect score is the detector genuinely finding the **injected** signature in real held-out subjects' recordings — not leakage. But the positives were manufactured, so **this measures detection of a planted fall, not of a real person falling.** §7.2 states the consequence.



One injected impact. A large, stereotyped, high-signal transient produced by a single generator — which is exactly why 0.9999 cannot be read as fall-detection performance.

4.4 Future-risk predictor (UniMiB SHAR, shipped)

Subject-disjoint 60/20/20: **18 / 6 / 6** subjects; 31,122 / 9,591 / 10,415 windows. Validation subjects [12, 13, 15, 18, 24, 29]; test subjects [2, 3, 5, 9, 14, 28]. The test set was touched **once**, after every choice was frozen.

Model	Precision	Recall	F1	ROC-AUC	PR-AUC
Majority class	0.0000	0.0000	0.0000	0.5000	0.128565
Logistic regression	0.4760	0.8432	0.6085	0.904389	0.659633
Hist gradient boosting (**better, not shipped**)	**0.7916**	0.6721	**0.7270**	**0.945913**	**0.815000**
Random forest, raw vote @ 0.5	0.6534	0.7827	0.7122	0.936848	0.764569
Random forest, calibrated @ HIGH ≥ 0.314046 (shipped)	**0.6707**	**0.7498**	**0.7080**	**0.936848**	**0.764569**

Confusion matrix at the HIGH threshold: **TN 8,583 · FP 493 · FN 335 · TP 1,004**. Accuracy **0.9205**; Brier 0.056675; specificity 0.9457.

Calibration — Platt scaling fitted on validation only (a = 1.24940294, b = −1.07904853). Brier improves from 0.062600 (raw) to **0.056675** (calibrated). The top bin is close (0.9–1.0: 496 windows, predicted 0.9407, observed 0.8952), but the **middle bins are over-confident** — the 0.3–0.4 bin predicted 0.3461 and was positive 0.5000 of the time. The probability is shown in the app *only* because it is calibrated, and always alongside the band, thresholds and version.

Risk bands (fitted on validation, then frozen): HIGH ≥ **0.314046** (the max-F1 threshold); MODERATE ≥ **0.069201** (the largest threshold whose validation recall is still ≥ 0.95).

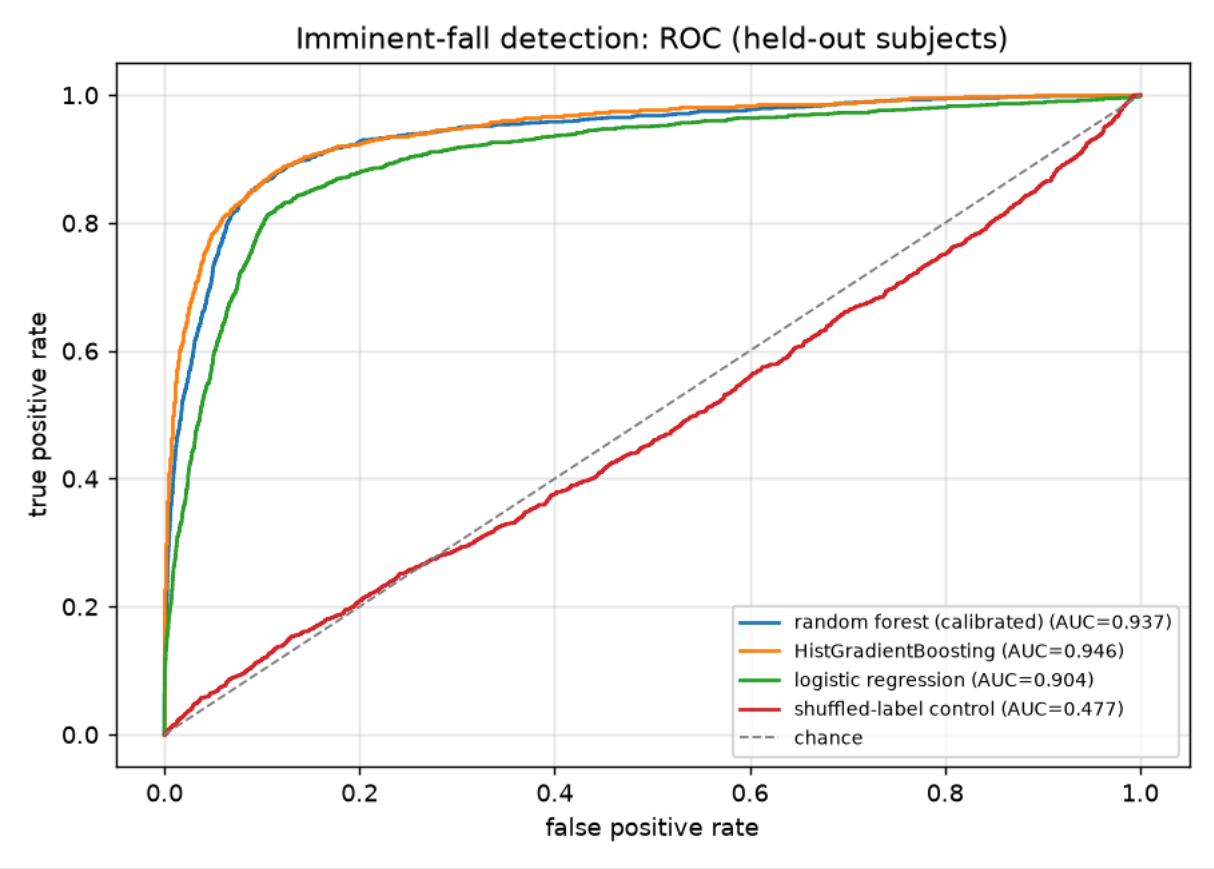
Detection across the horizon — it does not collapse as the lead grows, so there is real pre-impact signal:

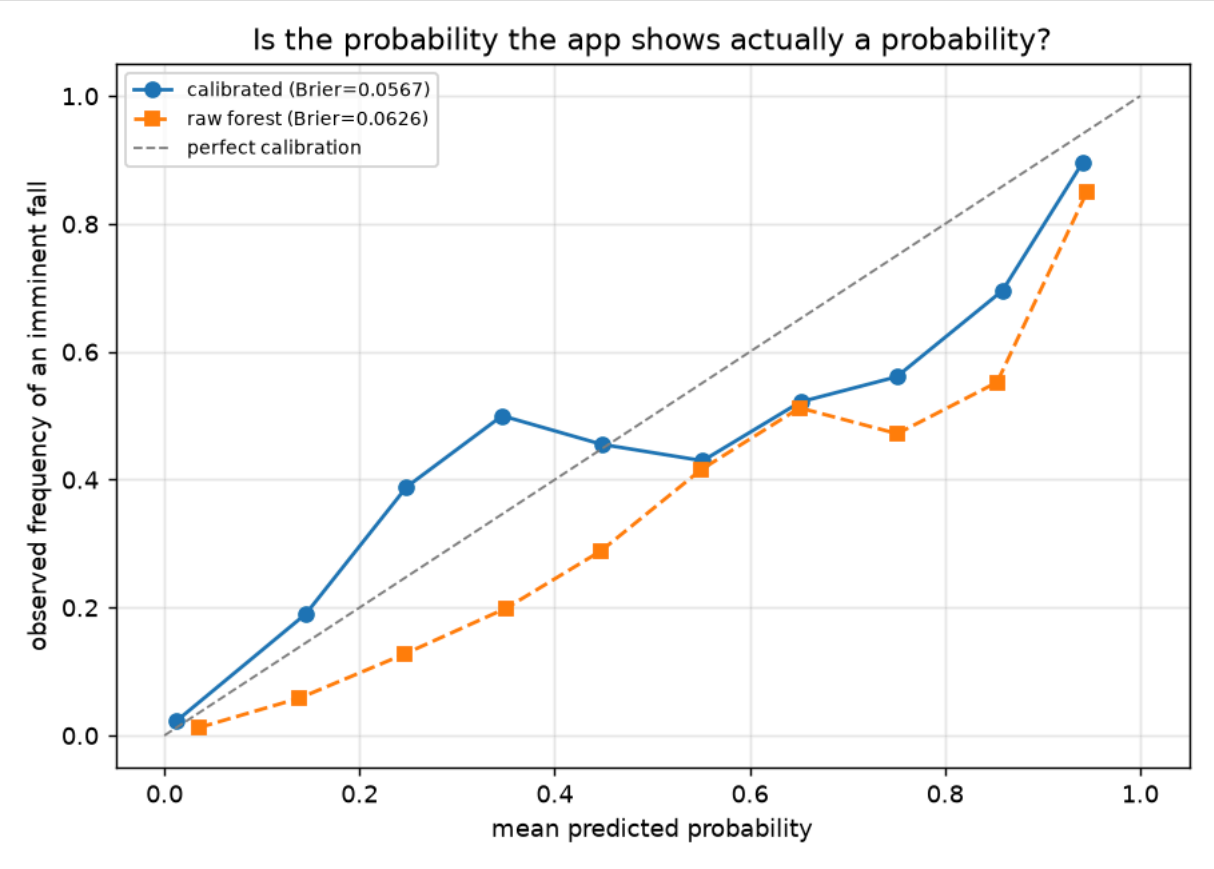
Lead	Positive windows	Detected	Rate
0.2 s	284	204	0.7183
0.4 s	281	199	0.7082
0.6 s	281	217	0.7722
0.8 s	270	213	0.7889
1.0 s	223	171	0.7668

Trial-level sensitivity — the number that matters most: of **284** held-out fall trials, **264** were flagged HIGH at least once → **0.9296**. This is the honest headline; window recall (0.7498) is the number underneath it.

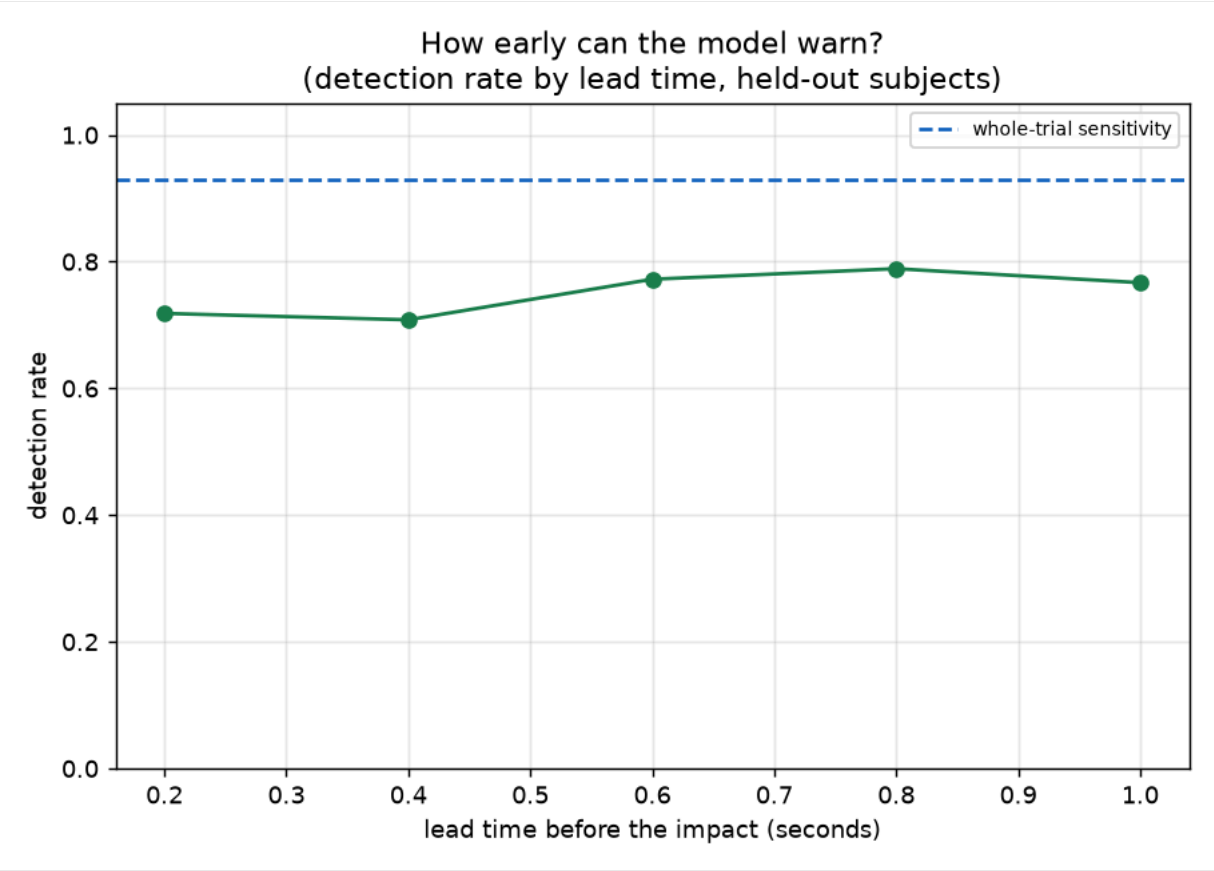
False alarms on ordinary activity: 493 of 9,076 ADL windows → **5.09 false alarms per minute** at the app's 0.64 s cadence (16.30/min at the raw 0.2 s stride). By activity: **jumping 16.1 %**, **sitting down 15.6 %**, **lying down 12.7 %**, going downstairs 4.4 %, standing up from floor 3.2 %, going upstairs 2.5 %, standing up from sofa 2.8 %, running 1.6 %, walking 1.1 %. The three worst confusions are the three movements where a body genuinely drops under gravity.

Leakage controls, checked rather than asserted: every positive window ends before its own impact (true); all leads within the horizon (true); the feature window never overlaps the label window (true); min/max lead 10/50 samples. Second control — the **shuffled-label run** scores ROC-AUC **0.47743** / PR-AUC **0.129885**, i.e. chance.





ROC-AUC 0.9368 looks strong; the calibration plot is the more important one — the top bin tracks the diagonal but the mid-range sits above it.



Detection rate from 0.2 s to 1.0 s before impact. It holds up at the full second, which is the interesting positive result.

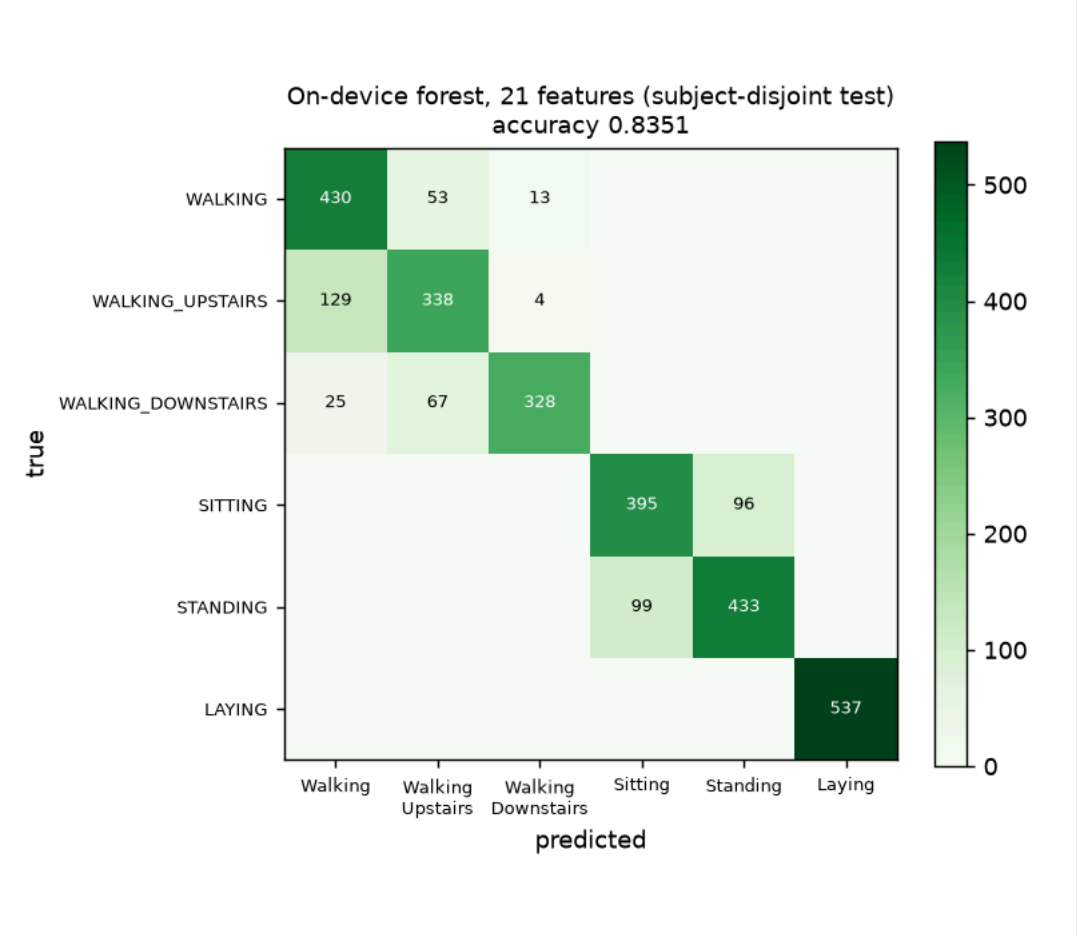
4.5 Transport-mode classifier (GeoLife labels)

Explicitly **not** a risk model — it predicts the next segment's transportation mode.

9,584 labelled segments from 69 users; subject-disjoint split 45 / 19 users (8,036 / 1,548 segments); `RandomForest`, 300 trees, 9 features, balanced class weights.

Accuracy 0.8475 · macro-F1 0.5784 · weighted-F1 0.8507.

Per-class on unseen users: walk 0.951 F1 (827) · bike 0.829 (168) · bus 0.818 (292) · subway 0.690 (117) · car 0.588 (77) · train 0.545 (11) · taxi 0.362 (53) · motorcycle 1.000 (1) · boat, run 0.000 (1 each). The macro-F1 is the mean over the **10** classes with test support (`airplane` has support 0). Top features: `p95_speed_mps` 0.200, `distance_m` 0.186, `mean_speed_mps` 0.160, `straightness` 0.117.



The shipped 24-feature activity model's confusion matrix: losses are almost entirely inside the walking triad; sitting, standing and laying hold up.

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5. Training methodology (compact)

	Trajectory anomaly	Activity	Impact detector	Future-risk
Data	GeoLife	UCI HAR	UCI HAR + injected impacts	UniMiB SHAR
Target	none (unsupervised)	6 real activity labels	injected impact vs untouched	`IMMINENT_FAL`
Features	14 movement features	561 (reference) / **24** (shipped)	24	**22** acceleromo
Window	point-level + 5-point windows	128 samples @ 50 Hz (2.56 s)	128 samples	50 samples (1
Horizon	none (not a forecast)	none	none	**1.0 s**
Split	`traj_uid` 900 / 300 traj, 0 overlap	official subject-disjoint 21 / 9	subject-disjoint 21 / 9	subject-disjoi
Preprocessing	standardise; segment at gaps > 120 s	dataset-provided; m/s² → g	same 24 features	**none that ne** free per windo
Hyper-parameters	200 trees, contamination 0.02	300/561; **32 trees, depth 8**; 1 tree baseline	40 trees	**120 trees, de balanced_sub**
Balancing / threshold	quantile thresholds (0.98 → 2 %)	none needed	balanced by construction; app gate ≥ 0.55	`balanced_sub` MODERATE 0.
Seed	42	fixed	fixed	42
Selection basis	single configuration	**deployment size** (JSON asset walkable in Kotlin)	1 configuration + shuffled control	**2,000,000-by** of 65 configs

	Trajectory anomaly	Activity	Impact detector	Future-risk
Artifact	`ml/models/isolation_forest.joblib`	`app/src/main/assets/har_activity_model.json`	`app/src/main/assets/har_fall_model.json`	`app/src/main` (1,974,297 B,

Model selection under the size budget (future-risk). `tune.py` searched **65 configurations** on validation only. The winner by validation PR-AUC was rejected because it cannot ship: 300 trees unlimited depth → val PR-AUC 0.8759 but **20,059.7 KB** / 700,492 nodes. 300 trees depth 8 → 0.8201 but **2,003,677 bytes**, over the 2,000,000-byte budget by 3,677. The shipped 120-tree depth-11 forest → **0.8451 / 0.9709** in **1,968,314 bytes** / 66,894 nodes. The constraint costs **−0.031 validation PR-AUC**, and it is the reason the shipped model is not the best this pipeline can fit. The rule is stored in `tuning.json` and re-derived by a test.

Training-and-deployment parity (verified, not assumed).

Model	Max feature error	Max probability error	Argmax mismatches
Activity	5.458e-07	4.970e-07	0
Impact / fall	6.342e-07	4.736e-07	0
Future-risk	parity asserted against scikit-learn on **16 held-out windows** (features, forest and calibration)		

A test also asserts the app asset is **byte-identical** to the verified training output, so the shipped model cannot drift from the trained one.

Environment: Python **3.11.9** (Windows); numpy 2.4.4, pandas 3.0.6, scipy 1.17.1, **scikit-learn 1.8.0**, matplotlib 3.11.2, pytest 9.1.1, joblib 1.5.3. Seed **42** everywhere. A single desktop with ~8 GB RAM, no GPU. Recorded runtimes: preprocess 539.2 s · EDA 56.5 s · anomaly train 76.7 s · anomaly evaluate 74.01 s · HAR EDA 12.9 s · history 34.1 s · OSM 10.8 s · weather 10.6 s · transport 282.2 s · future-risk train ~4 min · future-risk tuning ~10 min.

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6. Four risk concepts, kept separate

Shown as	What it actually is	Trained?	Has a probability?
Geofence status	A configured rule: inside/outside a polygon the authority drew	No	No
Approach risk	A published weighted checklist over crossing history	No — documented heuristic	No
Activity	A 6-class classifier over the current 2.56 s window	Yes	Confidence only
Movement / future risk	An impact detector and a 1-second-ahead impact forecast	Yes	Yes, calibrated (future-risk)

Enforced in code and tests: distinct types throughout, so a screen cannot render the heuristic under the trained model's heading; the card's own text names the other concepts and denies being them (asserted by a test); **risk never triggers SOS** — `backend/tsm/api.py` ignores risk entirely when computing state, and a HIGH prediction leaves `latestSos` empty and `openSosCount == 0` (asserted by `backend/tests/test_risk.py`). Predictions live in their own table, their own endpoints, and their own dashboard panel — never the SOS queue or the incident feed.

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7. Consolidated results and the limits of this evidence

7.1 Consolidated results

Model / component	Dataset	Evaluation split	Sample count	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Evidence source
Trajectory anomaly (unsupervised)	GeoLife 1.3	trajectory-disjoint	369,247 pts / 300 traj (17,462 injected)	—	0.6165 (P@k)	0.6165 (R@k)	—	0.8922 (injection)	0.6763	`ml/outputs/eval/eval_report.json`
Activity — 561-feature reference	UCI HAR	subject-disjoint	2,947 windows / 9 subjects	0.9287	0.9301 (macro)	0.9258 (macro)	0.9271	—	—	`ml/outputs/har/train/har_train_re`

Model / component	Dataset	Evaluation split	Sample count	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Evidence source
Activity — 24-feature on-device (shipped)	UCI HAR	same	2,947 windows	**0.8351**	0.8405 (macro)	0.8307 (macro)	**0.8333**	—	—	same
Impact detector (shipped, synthetic positives)	UCI HAR + injected	subject-disjoint	1,848 windows (924/924)	0.9962	0.9957	0.9968	**0.9962**	**0.9999**	0.9999	`ml/outputs/har/fall/har_fall_repo`
Future-risk — majority class	UniMiB SHAR	subject-disjoint	10,415 windows / 6 subjects	0.8714	0.0000	0.0000	0.0000	0.5000	0.128565	`ml/outputs/future_risk/metrics.js`
Future-risk — hist gradient boosting (**not shipped**)	UniMiB SHAR	same	10,415	—	0.7916	0.6721	0.7270	0.945913	**0.815000**	same
Future-risk — calibrated @ HIGH (shipped)	UniMiB SHAR	same	10,415 (1,339 pos)	**0.9205**	0.6707	**0.7498**	0.7080	**0.936848**	0.764569	same
Transport-mode classifier	GeoLife `labels.txt`	subject-disjoint	1,548 segments / 19 users	0.8475	0.5922 (macro)	0.5775 (macro)	0.5784 (macro) / 0.8507 (weighted)	—	—	`ml/outputs/transport/model_metric`

Event-level: future-risk **trial sensitivity 0.9296 (264/284)**; trajectory anomaly **group overlap between train and test = 0**.

7.2 Limits of this evidence

These are properties of what **was** built, so they belong in the report rather than being cut away.

1. **Because the impact detector's positives are synthetic, its 0.9999 is not a fall-detection claim.** It measures detection of an injected signature.
2. **The future-risk model raises about 5.09 false alarms per minute of ordinary activity**, worst on jumping (16 %), sitting down (16 %) and lying down (13 %), and its mid-range probabilities are over-confident. It is an informational band plus a queued summary — **never** an alarm, and it never sends an SOS.
3. **The label is a fall impact, not a tourist incident.** UniMiB is 30 healthy adults in a laboratory; UCI HAR is a scripted waist-mounted protocol. No real incident, injury, crime or medical outcome exists in any dataset used, so no supervised tourist-risk model is possible from this data.
4. **The model cannot tell a dropped phone from a falling person** — no phone-drop class exists in any dataset used.
5. **Nothing here has run on a physical phone or emulator,** and the app's screens, sensors, GPS and HTTP stack have therefore not been exercised on hardware.
6. **No user study, interview, survey or usability evaluation was conducted** at any point.
7. **Some metrics do not exist and are not estimated:** real-anomaly accuracy/F1 for the unsupervised model (no labels); ROC-AUC, PR-AUC, per-class specificity and calibration for the HAR and transport models (never saved); repeated cross-validated subject-disjoint evaluation (single fold only); the impact detector's training-window count.
8. **Public and laboratory datasets do not validate real tourist safety.** Every metric here is a statement about models on held-out laboratory and public data — not about a tourist.

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8. Evidence appendix

8.1 Model artifacts

Artifact	Bytes	Version	Used by
`ml/models/isolation_forest.joblib`	1,753.5 KB	—	offline analysis
`app/src/main/assets/har_activity_model.json`	268,734	—	shipped, on-device
`app/src/main/assets/har_fall_model.json`	59,679	—	shipped, on-device

Artifact	Bytes	Version	Used by
`app/src/main/assets/har_activity_model_spec.json`	3,479	—	shipped documentation asset
`app/src/main/assets/future_risk_model.json`	**1,974,297**	`future-risk-unimib-v1`	shipped, on-device
`app/src/test/resources/future_risk_reference_cases.json`	28.6 KB	—	Kotlin parity test (16 windows)
`app/src/test/resources/har_reference_cases.json`	199.1 KB	—	Kotlin parity test (12 windows)
`app/src/test/resources/har_fall_reference_cases.json`	295.6 KB	—	Kotlin parity test (12 windows)

8.2 Commands

```
# from the repository root (Windows, Python 3.11.9)
python ml\preprocess.py            # GeoLife parse + clean
python ml\eda.py                   # 12 figures + tables
python ml\train.py                 # anomaly model
python ml\evaluate.py              # injection evaluation
python ml\har_preprocess.py        # UCI HAR extract + verify
python ml\har_train.py             # activity model
python ml\har_fall.py              # injected-impact detector
python ml\har_verify_export.py     # export vs scikit-learn: expect ALL PASS
python ml\har_eda.py
python ml\history.py
python ml\osm_eda.py               # cache; --refresh to re-fetch
python ml\weather_eda.py           # Open-Meteo unless OPENWEATHER_API_KEY is set
python ml\transport_mode.py
python ml\future_risk\train.py     # future-risk model + app asset
python ml\future_risk\tune.py      # the 65-config size-budget search

powershell -NoProfile -ExecutionPolicy Bypass -File ml\run_all.ps1   # everything offline
# -Network for the two fetches, -Transport and -FutureRisk for the two slow models

python -m pytest ml\tests ml\future_risk\tests -q
```

8.3 Test counts (measured, this round)

Suite	Command	Executed	Passed	Failed	Errors	Skipped
Android JVM unit	`:app:testDebugUnitTest`	**548**	**541**	0	0	7
Backend (pytest, in-process)	`python -m pytest backend/tests -q`	**265**	**265**	0	0	0
Backend acceptance (real subprocess)	`python backend/verify_integration.py`	**41**	**41**	0	0	0
Python / ML (pytest)	`python -m pytest ml/tests ml/future_risk/tests -q`	**179**	**179**	0	0	0
Android instrumented	`:app:compileDebugAndroidTestKotlin`	0 (compile only)	—	—	—	—

The **7 skipped** Android cases are the opt-in live-backend tests, which run only with `-Dtsm.e2e.baseUrl`. Run against a real Flask server they were **7/7 passed,** `skipped="0"`. The 179 ML tests are `ml/tests` (86) plus `ml/future_risk/tests` (93). Instrumented Android tests were never executed — no device or emulator existed.

8.4 Model artifacts, figures and raw outputs

- **GeoLife:** `ml/outputs/clean/` (`trajectory_summary.csv` 18,670 rows, `points_clean_sample.csv` 1,429,740 rows), `ml/outputs/eda/` (12 figures, 11 tables), `ml/outputs/train/`, `ml/outputs/eval/` (`eval_report.json`, `eval_metrics.csv`, `top_anomalies.csv`, `injection_scores.csv`), `ml/outputs/predict/`.
- **UCI HAR:** `ml/outputs/har/clean/`, `ml/outputs/har/eda/` (6 figures, 5 tables), `ml/outputs/har/train/` (reports, per-class, figures, model JSON), `ml/outputs/har/fall/` (ROC, confusion, signal figure, model JSON).
- **UniMiB / future-risk:** `ml/outputs/future_risk/` — `metrics.json`, `dataset_provenance.json`, `tuning.json`, `calibration_reliability.csv`, `test_predictions.csv` (10,415 rows), `future_risk_reference_cases.json`, `figures/` (6 plots).
- **Map / weather / history / transport:** `ml/outputs/osm/`, `ml/outputs/weather/`, `ml/outputs/history/`, `ml/outputs/transport/`.

Every number in this report was read out of a file in `ml/outputs/`, `ml/models/` or `app/src/main/assets/`. Where evidence does not exist, the text says so instead of supplying a number. The full-length version of this report, including the complete figure galleries, the dataset-by-dataset licence discussion and the full unavailable-metrics inventory, is preserved in `docs/data_and_ml_analysis_full.md`.